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The Blind Spot Paradox

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When a self-repairing model destroys the evidence its own monitor needs

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The Blind Spot Paradox. When a self-repairing model destroys the evidence its own monitor needs. Four students of the CentraleSupélec research track, supervised by Raphaël Minato.

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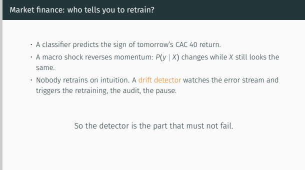
The Blind Spot Paradox

└─Market finance: who tells you to retrain?



We work in market finance. Take a classifier that predicts the sign of tomorrow's CAC 40 return. A macro shock reverses momentum: the conditional law of y given X changes while X still looks the same. In production nobody retrains on intuition. A drift detector watches the error stream and triggers the retraining, the audit, the pause.

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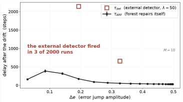
└The classifier erases its own alarm

The classifier erases its own alarm



Authors' experiment R2, re-run locally. $\lambda = 50$, 2 000 runs, 20 amplitudes × 100 seeds.

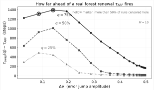
Modern models repair themselves. An adaptive random forest replaces its worst trees within a few dozen steps, and the error goes back down. The detector accumulates evidence on that error, and the evidence is gone before its threshold is reached. The classifier working correctly is what breaks the monitoring. We re-ran the authors' experiment. At threshold fifty, over two thousand runs, the detector fired three times. The red curve is missing because there is almost nothing to plot.

Two literatures, one missing clock

- **CUSUM** (Page, 1954): the alarm delay grows as the threshold divided by the size of the jump, $\tau^{*}_{\mathrm{det}} = \lambda/(\Delta e - \delta_P)$.
- **Adaptive Random Forest** (Gomes et al., 2017): evaluated on predictive accuracy alone.
- **Gama et al. 2014, Lu et al. 2018**, the two reference surveys: neither lists a measure of how long a model takes to recover.
- **Source paper**: on heteroscedastic streams, no admissible threshold exists. $\lambda \geq 15$ to avoid false alarms, $\lambda \leq 12.4$ to catch the drift.

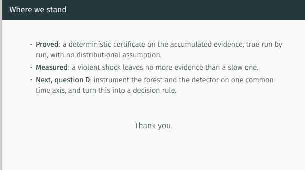
CUSUM, nineteen fifty-four: the alarm delay grows as the threshold divided by the size of the jump. Adaptive random forest, two thousand seventeen: evaluated on predictive accuracy alone. Gama and Lu, the two reference surveys, list detection delay and accuracy. Neither lists how long a model takes to recover. And the source paper reports that on heteroscedastic streams no admissible threshold exists: fifteen or above to avoid false alarms, twelve point four or below to catch the drift.

We measured the missing clock



τ_{ARF} dates the first tree out of ten. A real forest renewal comes 6.0 to 14.3 times later.

So we measured recovery, on our own instrumented campaign: two thousand runs, twenty amplitudes, one hundred seeds, ten trees, every tree replacement logged. The adaptation date used in the literature, tau ARF, is the first replacement: one tree out of ten. We compared it with the date at which a real fraction of the forest has been renewed. Tau ARF is always a lower bound. Zero violations on five thousand seven hundred and ninety-five uncensored comparisons, and this is a counting argument, not a statistical one: it holds run by run. For three quarters of the forest, the gap runs from one hundred sixty-seven to one thousand three hundred sixty-eight steps. The clock used to say a model has recovered is optimistic, always in the same direction.



Two more results are in the report: a deterministic certificate on the evidence, and the measured finding that a violent shock leaves no more evidence than a slow one. What remains is question D: instrumenting the forest and the detector on one common time axis, and turning this into a decision rule for the practitioner. Thank you.